PYTHON DATA PROCESSING

1. Introduction

Python Data Processing is a project that processes structured records and generates summary statistics. The project uses fictional student records to demonstrate how numerical information can be organized, analyzed, and presented in a meaningful format.

It combines Python concepts such as lists, dictionaries, functions, sorting, filtering, and built-in statistical operations.

2. Objective

The main objectives are:

- To process a collection of structured records.
- To calculate useful summary statistics.
- To create functions for repeated operations.
- To generate examples of processed records.
- To understand basic data-processing techniques using Python.

3. Problem Statement

Structured datasets often contain multiple records and numerical attributes. Extracting useful information manually can be time-consuming.

This project addresses the problem by processing student records automatically and generating statistics that help summarize the dataset.

4. Tools and Technologies Used

Python 3: Used to implement the data-processing program.

Lists and Dictionaries: Used to store student records and their attributes.

Functions: Used to organize reusable operations.

Statistics Module: Used to calculate mean and median.

Collections Module: Used to count records by department.

Visual Studio Code: Used to write and execute the program.

GitHub: Used to store source code and documentation.

5. Dataset Description

The project uses eight manually created fictional student records.

Each record includes:

- Student ID
- Name
- Department
- Age
- Marks

The dataset contains students from Computer Science, Data Science, Artificial Intelligence, and Electronics.

6. Methodology

The project follows these steps:

1. Create a structured collection of student records.
2. Display the original dataset.
3. Extract numerical values from the records.
4. Calculate summary statistics.
5. Count students by department.
6. Sort students according to marks.
7. Filter students who meet a marks threshold.
8. Display the processed results and overall summary.

7. Data Processing Operations

7.1 Numerical Statistics

The program calculates:

- **Count:** Number of records analyzed.
- **Total:** Sum of the numerical values.
- **Average:** Sum divided by the number of values.
- **Median:** Middle value of the sorted dataset.
- **Minimum:** Smallest value.
- **Maximum:** Largest value.
- **Range:** Difference between the maximum and minimum values.

7.2 Department Summary

The program counts the number of students in each department. This provides a simple summary of the distribution of student records.

7.3 Top Student Identification

The program sorts student records by marks in descending order and displays the top three students.

7.4 Record Filtering

The program filters students who have scored at least 80 marks. This demonstrates how conditions can be applied to structured records to select relevant information.

8. Sample Dataset

Student ID	Name	Department	Age	Marks
S101	Arun	Computer Science	20	85
S102	Priya	Data Science	21	92
S103	Rahul	AI	19	76
S104	Sneha	Computer Science	22	68
S105	Kiran	Data Science	20	81
S106	Ananya	Electronics	18	95
S107	Vikram	Computer Science	23	74
S108	Meera	AI	21	88

9. Results and Summary Statistics

The program produces the following results from the sample dataset:

Statistic	Result
Total student records	8
Total marks	659
Average marks	82.375
Median marks	83
Highest marks	95
Lowest marks	68
Marks range	27
Students scoring at least 80	5

The top three students by marks are Ananya, Priya, and Meera.

Program Output Screenshot: Insert a screenshot showing the program execution and summary statistics.

10. Testing and Validation

The program should be tested by checking the displayed records, numerical statistics, department counts, top three students, and filtered records.

The expected numerical results should be compared with the actual terminal output. Any mismatch should be investigated before the project is submitted.

11. Concepts Learned

This project provides practical experience in:

- Structured data representation
- Lists and dictionaries

- Functions
- Numerical calculations
- Mean and median
- Sorting and filtering
- Counting categorical values
- Standard-library modules
- Generating summary reports

12. Limitations and Future Improvements

The current program uses a small, manually created dataset. It does not import external files or provide a graphical interface.

Future improvements could include importing CSV and Excel files, using Pandas for larger datasets, generating charts, and exporting processed results to CSV.

13. Conclusion

The Python Data Processing project demonstrates how Python can transform structured records into useful summary information. It combines several fundamental programming concepts and provides a starting point for more advanced data analysis.

14. Project Links

GitHub Repository: https://github.com/venky1318/python-data-processing